Is $\|S\|_{L^\infty} \le C\|\omega\|_{L^\infty}$? A deep check of `PeriodicStrainLinfBound`

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Abstract

`PeriodicStrainLinfBound` asks for one constant C with $\|S(t,x)\| \le C\|\omega(t)\|_{L^\infty}$ for every smooth periodic Navier–Stokes solution. We show it is false as stated, already at $t = 0$: an explicit family of divergence-free trigonometric polynomials on $\mathbb{T}^3$ has $\|\omega_K\|_\infty \le 1.403$ and $\|S_K\|_\infty \ge \frac{2}{\pi}\log K$. The arithmetic divergence is proved in Lean; the Fourier identity and the Gibbs bound are classical and are checked numerically. We record the conditional results that do give the bound, and what remains of it for Navier–Stokes solutions.

1 Literature

Biot–Savart gives $\nabla u = \mathcal{R}\omega$ with $\mathcal{R}$ a matrix of Calderón–Zygmund operators (compositions of Riesz transforms). These are bounded on L^p, $1 < p < \infty$, and from L^∞ to BMO, but not on L^∞. For bounded vorticity the velocity is only log-Lipschitz (Yudovich; see Majda–Bertozzi, *Vorticity and Incompressible Flow*, Ch. 8). The standard sharp example is the Bahouri–Chemin steady state $\omega = \operatorname{sgn} x \operatorname{sgn} y$ on $\mathbb{T}^2$, whose velocity gradient diverges logarithmically at the origin (Bahouri–Chemin 1994; used by Denisov for double-exponential growth of $\nabla\omega$ in 2D Euler, and by Elgindi–Jeong in 3D symmetric settings). The quantitative substitute is the log inequality of Beale–Kato–Majda and Kozono–Taniuchi, $\|\nabla u\|_\infty \le C(1 + \|\omega\|_{\mathrm{BMO}}(1 + \log^+ \|u\|_{H^s}))$.

2 An explicit periodic counterexample family

Let $f_K(x) = \frac{4}{\pi}\sum_{k\ \mathrm{odd}\le K} \sin(kx)/k$, the Dirichlet partial sum of $\operatorname{sgn}(\sin x)$. On $\mathbb{T}^3$ put

$$\omega_K = f_K(x) f_K(y)\, e_z, \qquad \Delta\psi_K = f_K f_K, \qquad u_K = (-\partial_y\psi_K,\ \partial_x\psi_K,\ 0).$$

Then u_K is a divergence-free trigonometric polynomial with $\nabla\times u_K = \omega_K$. With $\omega_K = \sum A_{kl}\sin kx \sin ly$, $A_{kl} = 16/(\pi^2 kl)$, we have $\psi_K = -\sum A_{kl}\sin kx\sin ly/(k^2+l^2)$ and

$$S_{11}(0) = -\partial_{xy}\psi_K(0) = \frac{16}{\pi^2}\sum_{k,l\ \mathrm{odd}\le K} \frac{1}{k^2+l^2}, \qquad S_{12}(0) = 0,$$

so $\|S_K(0)\|_{op} = S_{11}(0)$.

Theorem 1 (`bcSum_unbounded`, proved in Lean)**.** $\sum_{i,j<n} \frac{1}{(2i+1)^2+(2j+1)^2} \to \infty$. *More precisely it is at least* $\frac{1}{8}\sum_{i<n} \frac{1}{i+1}$.

The Gibbs bound $\sup|f_K| \le \frac{2}{\pi}\,\mathrm{Si}(\pi) \approx 1.1790$, uniform in K, is classical; so $\|\omega_K\|_\infty \lesssim 1.39$ while $\|S_K\|_\infty \ge S_{11}(0) \sim \frac{2}{\pi}\log K$. The ratio $\|S\|_\infty/\|\omega\|_\infty$ is invariant under $u \mapsto \lambda u(\mu\,\cdot)$, so no rescaling removes the divergence.

Numerics (complete; exact sums at the origin, grid sup of $\|S\|_{op}$ on an $8K \times 8K$ grid up to $K = 255$).

K	$\|\omega_K\|_\infty$	$S_{11}(0)$	$\sup_x \|S\|_{op}$	$S_{11}(0) - \frac{2}{\pi}\log K$
7	1.4024	1.6585	1.6585	0.4196
15	1.3931	2.0977	2.0977	0.3737
31	1.3908	2.5385	2.5385	0.3523
63	1.3902	2.9796	2.9796	0.3420
127	1.3900	3.4209	3.4209	0.3370
255	1.3900	3.8621	3.8621	0.3345
511	1.3900	4.3034	–	0.3332
1023	1.3900	4.7447	–	0.3326

K	t	M	$\|S\|_\infty$	ratio
31	0.00	1.3908	2.5385	1.8252
31	0.20	1.0757	2.2023	2.0473
31	0.40	1.0199	1.9882	1.9494
31	0.60	1.0075	1.8391	1.8253
31	0.80	1.0042	1.7235	1.7163
31	1.00	1.0023	1.6293	1.6255
127	0.00	1.3899	3.4209	2.4612
127	0.20	1.0000	2.2628	2.2628
127	0.40	1.0000	2.0156	2.0156
127	0.60	1.0000	1.8563	1.8563
127	0.80	1.0000	1.7350	1.7350
127	1.00	1.0000	1.6371	1.6371

The first table shows $\|\omega_K\|_\infty \to 1.3900 = (\frac{2}{\pi}\,\mathrm{Si}\,\pi)^2$, the sup of $\|S\|$ attained at the origin, and $S_{11}(0) - \frac{2}{\pi}\log K \to\approx 0.332$: growth exactly $\frac{2}{\pi}\log K$. The second table evolves the $K = 31$ and $K = 127$ data by 2D Navier–Stokes (= z-independent 3D NS), $\nu = 0.005$, to $t = 1$: the ratio is largest near $t = 0$ and is then quickly smoothed (to ≈ 1.63 by $t = 1$).

3 Conditional results that do give the bound

- **Dini continuity** (classical CZ theory): if $\omega(t)$ has modulus of continuity ϖ with $\int_0^1 \varpi(r)\,dr/r < \infty$, then $\|\nabla u\|_\infty \le C\big(\|\omega\|_\infty + \int_0^1 \varpi(r)\frac{dr}{r} + \|\omega\|_{L^2}\big)$. This includes Hölder and log-Hölder moduli $\varpi(r) \lesssim |\log r|^{-a}$, $a > 1$. The constant depends on the modulus, not on $\|\omega\|_\infty$ alone; this is a different statement.
- **Log inequality** (BKM, Kozono–Taniuchi): bounded by $\|\omega\|_\infty$ or $\|\omega\|_{\mathrm{BMO}}$ times log of a higher norm. This is the right replacement and is what Section 2 of the cubic-source note now uses.

- **Direction regularity** (Constantin–Fefferman): Lipschitz (or $\frac{1}{2}$-Hölder, Beirão da Veiga–Berselli) continuity of $\xi = \omega/|\omega|$ on the high-vorticity set controls the stretching $\xi \cdot S\xi$ through a local, cancelling kernel. It does not give $\|S\|_\infty \le C\|\omega\|_\infty$; the family above is two-dimensional, with constant direction $\xi = e_z$, and still violates the bound.

4 Could it hold for Navier–Stokes solutions?

1. **Uniform constant: false.** `PeriodicStrainLinfBound` quantifies over all smooth periodic data with one C, at all $t \ge 0$ including $t = 0$. Each u_K is smooth periodic divergence-free data with a smooth local solution (classical local well-posedness; it is in fact global, being 2D). At $t = 0$ the bound fails as $K \to \infty$. In Lean this is not yet a formal refutation: the repo has no local-existence theorem to produce `IsPeriodicMaximalSolution` for u_K, so the Prop remains an unassumed named hypothesis, now recorded as false modulo classical local existence.

2. **Constant depending on the solution, up to T^*: open, and not enough.** Any smooth solution satisfies $\|S\|_\infty \le C_T\|\omega\|_\infty$ on each $[0, T]$, $T < T^*$, trivially. Whether C can be taken uniform up to a finite T^* is open. Even if it could, it gives only $M' \le CM^2$, a lifespan bound, not regularity (cubic-source note, §3).

3. **Positive-time smoothing.** Viscosity regularises the data instantly, so a bound with C depending on ν, $t > 0$ and the energy might hold away from $t = 0$ (the numerics are consistent with this). Near a putative singularity this is exactly what is unknown.

5 Record

Proved (Lean): `bcSum_unbounded`. **Classical, checked numerically:** Fourier identity for $S_{11}(0)$, Gibbs bound, $\frac{2}{\pi}\log K$ growth. **Conclusion:** `PeriodicStrainLinfBound` as stated is false (modulo local existence, not formalized). **Open:** solution-dependent constant up to T^*; not sufficient for regularity in any case.